

Responses to Reviewers and Cover letter

Manuscript ID: IPM-D-25-06425

Manuscript Title: ALD²: Adaptive Layer-wise Denoising Decoding for Hallucinations Mitigation in Large Vision-Language Models

Dear Editor-in-Chief,

We would like to express our sincere appreciation to all the reviewers and editors for their valuable time and effort given to reviewing our paper. Their constructive comments will greatly improve the quality of our paper. According to the suggestions of all reviewers, we have made corresponding revisions as detailed below.

Editor-in-Chief (comments and responses):

Thank you for submitting your manuscript to Information Processing and Management. I have completed my evaluation of your manuscript. The reviewers recommend reconsideration of your manuscript following major revision. I invite you to resubmit your manuscript after addressing the comments below. Please resubmit your revised manuscript by 02/16/2026.

When revising your manuscript, please carefully consider all issues mentioned in the reviewers' comments: please outline every change made in response to their comments and provide suitable rebuttals for any comments not addressed. Please note that your revised submission may need to be re-reviewed. To submit your revised manuscript, please log in as an author at <https://www.editorialmanager.com/ipm/>, and navigate to the "Submissions Needing Revision" folder. Information Processing and Management values your contribution and I look forward to receiving your revised manuscript.

Response: Thank you for your response and for coordinating the evaluation of our manuscript. We would like to express our gratitude for the opportunity to revise and resubmit our manuscript following the major revision recommendation. We have been dedicated to ensuring that all concerns raised by the reviewers are effectively addressed in the revised version of the manuscript. We have made every effort to provide detailed responses and implement necessary modifications to meet the expectations of the reviewers and improve the overall quality of the paper.

Associate Editor (comments and responses)

Thank you for submitting your paper to IP&M. We have reviews on your manuscript, and the reviewers are supportive of your research but also believe the paper would benefit from additional explanation and clarification of the main contributions. I hope you can complete the recommendation changes in a revision of your article.

Therefore, I invite you to respond to the comments and revise your manuscript. If you decide to resubmit, please include a detailed response to the reviewer highlighting your changes. Given the high volume of submissions to IP&M and the desire to maintain the journal's rapid processing time, a detailed response to

the reviewers aids significantly in this regard. To assist with this, here is one of many nice articles on crafting a detailed response to the reviewers

- <https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1005730#sec006>.

Response: We sincerely thank you for your thoughtful feedback and for coordinating the evaluation of our manuscript. We greatly appreciate the opportunity to revise and resubmit our work. In the revised paper, we have addressed all the reviewers' concerns. The specific revisions are as follows:

- ① We discussed the decoding latency in detail, explicitly analyzing the trade-off between computational cost and reliability in practical deployments. We clarified that this cost is justified in safety-critical scenarios and outlined optimization strategies using visual token pruning to mitigate overhead.
- ② We discussed the existing work related to hallucinations in LVLs in Section 5.1.
- ③ We carried out extra experiments on InternVL-3.5-8B to improve the comprehensiveness and practicality of our work.
- ④ We carried out extra ablation experiments to further clarify why multiplicative decoding along with visual token pruning is effective.

We are grateful for your constructive guidance and have made every effort to implement these revisions to improve the overall quality of our paper.

Please refer to our point-to-point responses to the reviewer's comments below for more details.

To All Reviewers:

We sincerely thank you for your detailed and professional review comments, which have provided valuable guidance for further improving the quality of our paper.

In response to your suggestions and questions, we have made corresponding revisions and refined the content of the manuscript accordingly. The following presents the specific modifications we have made:

Attention:

In the following document, black text denotes the your comments, **blue text** presents our responses, **gray text** highlights the relevant context from the manuscript, and **red text** indicates the corresponding revisions of the manuscript.

To Reviewer #1:

Comment #1.1:

The proposed lightweight predictor relies solely on textual prompts to predict the optimal denoising layer, which may limit its effectiveness in cases where hallucinations are strongly driven by visual ambiguity rather than linguistic structure.

Thank you for your comments. We have pointed out in section Limitations that ALD² can't fully resolve the issue that in some cases LVLs are fully confident in their responses. Indeed, when hallucinations are strongly driven by visual ambiguity, LVLs are fully confident and ALD² focuses on situations where the model could have answered correctly originally, but due to noise, it ended up generating an incorrect response. The revised contexts from the manuscript are as follows:

Limitations

...

- While ALD² mitigates hallucinations by reducing noise in shallow layers, it does not explicitly model hallucination mechanisms or perform hallucination detection or correction, particularly in long-form responses (Huang et al., 2025; Zheng et al., 2025). As a result, in certain cases where LVLs produce highly confident but incorrect predictions, residual hallucination effects may still exist, despite being alleviated by ALD².

Data availability

The data and code will be released after the paper's publication.

Comment #1.2:

The quantification strategy for selecting the optimal denoising layer is token-level and supervised by ground-truth answers, which may not fully reflect realistic open-ended generation scenarios where such supervision is unavailable.

Thank you for your feedback. The quantification strategy for selecting the optimal denoising layer is to get more precise denoised layer during inference and in section 4.3 Ablation, our experiments demonstrate that in most cases randomly selecting a denoised layer outperforms the greedy decoding strategy. Additionally, when evaluating on CHAIR metrics, the predictor is trained on image captions which can reflect realistic open-ended generation scenarios. The relevant contexts from the manuscript are as follows:

Table 3

The importance of layer predictor. The numbers in parentheses indicate the change in accuracy when using random layer selection compared to using the predictor. The results in **bold** represent outcomes that exceed the greedy decoding results.

	<i>Random</i> ↑	<i>Popular</i> ↑	<i>Adversarial</i> ↑	CHAIR(C_s) ↓	CHAIR(C_f) ↓	MME score ↑
LLaVA-1.5-7B	81.17 (-3.66)	79.33 (-3.84)	77.50 (-3.33)	79.80 (+34.8)	34.50 (+21.5)	1116.15 (-531.35)
InstructBLIP-vicuna-7B	85.83 (-2.00)	80.17 (-1.16)	78.50 (-1.50)	64.0 (+21.80)	18.7 (+7.20)	1593.33 (-16.86)
LLaVA-Next-8B	89.17 (-0.33)	87.17 (-2.33)	84.50 (-0.83)	30.40 (+3.60)	8.30 (+1.70)	1707.05 (-151.39)

Comment #1.3:

The method introduces a non-negligible inference latency overhead, and while the paper reports this cost, there is limited discussion on how ALD² would scale to longer responses or higher-resolution visual inputs.

We would like to express our gratitude for the valuable feedback. We add the discussion on how ALD² would scale to longer responses or higher-resolution visual inputs in section 4.3 Ablation. The revised contexts are as follows:

4.3. Ablation

Latency.

..... Moreover, compared with greedy decoding, ALD2 performs additional inference over one quarter to all layers after the main decoding step, which we identify as the primary source of the increased latency. Regarding scalability, while our evaluation focuses on standard-length text responses and moderate-resolution visual inputs, the relevant latency overhead of ALD² may grow for longer responses or higher-resolution images. Since both the additional layer-wise inference and token processing scale with the number of tokens and visual features, scenarios with extended dialogues or dense visual inputs could experience further latency increases. Notably, the latency can be reduced by leveraging our visual token pruning strategy, where less informative tokens are directly removed from the computation graph rather than merely zeroing their hidden states. This reduces memory usage and computation in subsequent layers, offering a practical avenue to mitigate latency while retaining most of the accuracy improvements.

Effect of Multiplicative Decoding and Visual Token Pruning.

Comment #1.4:

The visual token pruning ratio is fixed across models and tasks, which may not be optimal in diverse scenarios, especially for complex scenes with fine-grained visual details.

Thank you for your valuable feedback. We have already discussed this issue in section Limitations. Our method primarily focuses on mitigating the noise introduced by shallow layers in LVLMs, and on helping LVLMs make correct predictions that they are inherently capable of producing but may fail to due to such noise. Therefore, for visual token pruning, we adopt a simple yet practical strategy by removing visual tokens with low attention scores at a fixed ratio, which helps us obtain a denoised representation distribution. The revised contexts are as follows:

Limitations

While the Adaptive Layer-wise Denoising Decoding (ALD2) method has significantly mitigated hallucinations and improved the reasoning reliability of Large Vision-Language Models by dynamically suppressing noise in shallow layers and emphasizing critical signals in deeper layers, it still has several limitations:

- ALD² denoises shallow layers by pruning a fixed proportion of visual tokens based on attention scores. This design choice favors simplicity and model-agnostic deployment, but it does not explicitly account for more complex vision-aware internal dynamics, such as adaptive token selection, vision-aware head behaviors, task-specific hallucination mitigation, or in-context visual reasoning (Zhou et al., 2025a; He et al., 2025; Zhou et al., 2024b). More precise token pruning strategies or dynamically adjusted pruning ratios may further improve denoising effectiveness. For example, Huo et al. (2024) proposed Self-Introspective Decoding (SID), which dynamically evaluates the importance of vision tokens during decoding to adaptively reduce hallucinations in large vision-language models.

Comment #1.5:

The evaluation focuses primarily on object-level hallucinations and general benchmarks, while other forms of hallucinations such as relational, temporal, or causal hallucinations are not explicitly analyzed.

Thank you for your comments. ALD² focuses on single image2text task so that there are no temporal hallucinations in such task and the related works are also evaluated on these datasets and benchmarks. We view this as a promising direction for future work and believe that ALD² could be extended to broader domains, beyond the current LVLM setting, by adapting the layer-wise denoising principle to other multimodal or generative models.

Comment #1.6:

The paper does not sufficiently position ALD² with respect to recent work on hallucination behavior in long-form responses and vision-aware internal dynamics, which would help clarify the novelty and limitations of the approach. The discussion of limitations should explicitly acknowledge and cite the following relevant works to better contextualize the proposed approach:

- * Why LVLMs Are More Prone to Hallucinations in Longer Responses
- * Visual In-Context Learning for Large Vision-Language Models
- * Cracking the Code of Hallucination in LVLMs with Vision-aware Head Divergence
- * Improving Medical Large Vision-Language Models with Abnormal-Aware Feedback
- * Visual hallucination detection in large vision-language models

Thank you for your suggestion. We agree that a clearer positioning of ALD² with respect to recent studies on hallucination behavior and vision-aware internal dynamics would help better contextualize both the novelty and the limitations of our approach. Our work differs from these prior studies in that ALD² does not aim to explicitly model hallucination mechanisms, detect hallucinations, or design task-specific feedback signals. Instead, ALD² focuses on reducing noise introduced by shallow visual representations in LVLMs through a lightweight attention-based visual token pruning strategy, with the goal of improving the confidence of predictions that the model is already capable of producing. The cited works investigate complementary aspects of LVLM behavior, including hallucination amplification in long-form responses, vision-aware internal dynamics, in-context visual learning, and task-specific hallucination mitigation strategies (e.g., medical feedback or hallucination detection). We explicitly acknowledge these differences and discuss the limitations of ALD² in the Limitations section, clarifying that our approach does not directly address hallucination detection or long-form reasoning behaviors. The revised section Limitations are as follows:

Limitations

While the Adaptive Layer-wise Denoising Decoding (ALD²) method has significantly mitigated hallucinations and improved the reasoning reliability of Large Vision-Language Models by dynamically suppressing noise in shallow layers and emphasizing critical signals in deeper layers, it still has several limitations:

- ALD² denoises shallow layers by pruning a fixed proportion of visual tokens based on attention scores. This design choice favors simplicity and model-agnostic deployment, but it does not explicitly account for more complex vision-aware internal dynamics, such as adaptive token selection, vision-aware head behaviors, task-specific hallucination mitigation, or in-context visual reasoning (Zhou et al., 2025a; He et al., 2025; Zhou et al., 2024b). More precise token pruning strategies or dynamically adjusted pruning ratios may further improve denoising effectiveness. For example, Huo et al. (2024) proposed Self-Introspective Decoding (SID), which dynamically evaluates the importance of vision tokens during decoding to adaptively reduce hallucinations in large vision-language models.
- ALD² requires a training phase prior to inference. Although the predictor is lightweight, it still introduces additional computational overhead and latency. Our experiments demonstrate that this overhead is non-negligible, which may limit the applicability of ALD² in latency-sensitive scenarios.
- While ALD² mitigates hallucinations by reducing noise in shallow layers, it does not explicitly model hallucination mechanisms or perform hallucination detection or correction, particularly in long-form responses (Huang et al., 2025; Zheng et al., 2025). As a result, in certain cases where LVLMs produce highly confident but incorrect predictions, residual hallucination effects may still exist, despite being alleviated by ALD².

To Reviewer #2:

Summary:

This paper proposes ALD² (Adaptive Layer-wise Denoising Decoding), a decoding-time framework to mitigate hallucinations in large vision-language models (LVLMs). The authors analyze hallucinations from a layer-wise noise propagation perspective, arguing that redundant visual tokens in shallow layers introduce noise that later overrides correct token predictions. To address this, the paper combines visual token pruning at adaptively selected shallow layers with multiplicative decoding, guided by a lightweight layer predictor. Extensive experiments on CHAIR, POPE, and MME benchmarks demonstrate consistent improvements across multiple LVLM backbones.

Strength:

- The layer-wise noise interpretation of hallucinations is insightful and complements existing contrastive decoding and post-hoc correction methods.
- The combination of shallow-layer denoising and multiplicative decoding is intuitive and technically grounded. The adaptive layer selection further strengthens the approach.
- Results on CHAIR, POPE, and MME across several LVLMs are convincing and show consistent gains over strong baselines (Greedy, DoLa, ICD, VCD).

We are glad that you found our interpretation of layer-wise noise propagation insightful and recognized the intuitiveness and technical grounding of combining shallow-layer denoising with multiplicative decoding. Your acknowledgement of our experimental results on CHAIR, POPE, and MME benchmarks is greatly encouraging. Thank you for your time and constructive feedback.

Weakness:

Comment #2.1:

The decoding latency is significantly higher than greedy decoding, and the paper could better discuss when this trade-off is acceptable in real-world deployments.

We appreciate you raising the critical issue regarding decoding latency. As you rightly pointed out, discussing the trade-off between performance gains and computational overhead is essential for evaluating the real-world deployment of ALD². We add the discussion about this trade-off in section 6 Theoretical and Practical Implications as follows:

6. Theoretical and Practical Implications

...

From a practical perspective, the proposed ALD² framework offers a lightweight yet effective solution for reducing hallucinations in LVLMs, operating entirely at the decoding stage and easily integrating into existing pipelines. Its adaptive layer selection mechanism enables dynamic noise suppression tailored to different inputs, making it suitable for real-world deployment scenarios where reliability is critical. In practice, we believe the trade-off between increased decoding latency and improved trustworthiness is acceptable in settings where hallucination reduction and reasoning reliability are prioritized over strict efficiency constraints, such as offline evaluation, safety-critical decision support, medical or scientific analysis, and high-stakes human-in-the-loop applications. Conversely, for latency-sensitive scenarios like real-time interactive systems, greedy decoding or other lightweight strategies may be preferable. These considerations underscore that layer-wise denoising during decoding is both theoretically grounded and practically effective.

Comment #2.2:

As acknowledged by the authors, ALD² cannot eliminate hallucinations when models are highly confident in incorrect predictions.

We are grateful for highlighting this point. We have pointed out this question in section Limitations and ALD² focuses on situations where the LVLMs could have answered correctly originally, but due to the shallow layer noise, it ended up generating an incorrect response.

Comment #2.3:

I recommend including evaluations on more recent and stronger LVLMs (e.g., Qwen-VL-2.5) to better demonstrate the generality and scalability of the proposed method.

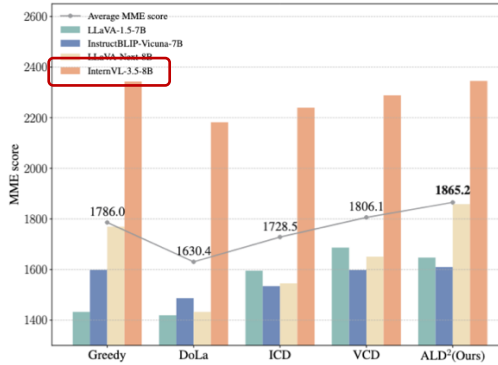
This is a highly constructive suggestion, and we appreciate it. We have conducted additional experiments on InternVL-8B, a state-of-the-art model comparable to or exceeding Qwen-VL-2.5 in performance, to address your concerns. For InternVL-3.5-8B model, ALD² outperforms baselines. The results are as follows:

Table 1
Overall performance on CHAIR metrics across multiple LVLMs. **Bold numbers** indicate the best results in each column.

Method	LLaVA-1.5		InstructBLIP		LLaVA-Next		InternVL-3.5	
	$C_S \downarrow$	$C_I \downarrow$	$C_S \downarrow$	$C_I \downarrow$	$C_S \downarrow$	$C_I \downarrow$	$C_S \downarrow$	$C_I \downarrow$
Greedy	51.0	14.4	42.6	11.8	33.2	7.7	32.0	9.7
DoLa	47.8	14.6	43.0	11.7	35.6	8.8	30.8	10.1
ICD	47.2	13.8	55.6	17.5	39.2	10.1	39.4	11.2
VCD	50.8	13.5	50.2	14.2	47.2	10.7	40.4	10.3
ALD ² (Ours)	45.0	13.0	42.2	11.5	26.8	6.6	28.2	8.1

Table 2
Overall performance on POPE metrics across multiple LVLMs. **Bold numbers** indicate the best results in each column.

Model	Method	Random		Popular		Adversarial		Average	
		F1 $\uparrow$	Acc. $\uparrow$	F1 $\uparrow$	Acc. $\uparrow$	F1 $\uparrow$	Acc. $\uparrow$	F1 $\uparrow$	Acc. $\uparrow$
LLaVA-1.5-7B	Greedy	79.40	79.50	77.20	76.67	76.58	75.83	77.73	77.33
	DoLa	78.35	78.16	76.33	75.50	76.21	75.33	76.93	76.33
	ICD	83.79	84.33	82.05	82.50	79.20	79.17	81.68	82.00
	VCD	84.83	85.33	79.44	80.50	77.40	78.00	80.56	81.28
	ALD ² (Ours)	83.24	84.83	81.74	83.17	79.64	80.83	81.54	82.94
InstructBLIP-Vicuna-7B	Greedy	83.33	83.33	76.34	75.00	76.52	74.00	78.73	77.44
	DoLa	82.59	84.33	77.30	78.16	76.63	77.33	78.84	79.94
	ICD	83.97	84.67	79.80	79.67	77.91	78.83	80.56	81.06
	VCD	84.43	85.00	76.97	75.67	75.19	73.17	78.86	77.95
	ALD ² (Ours)	87.08	87.83	81.27	81.33	80.39	80.00	82.91	83.05
LLaVA-Next-8B	Greedy	86.87	87.50	84.07	84.33	82.26	82.17	84.40	84.67
	DoLa	86.87	87.50	84.07	84.33	82.26	82.17	84.40	84.67
	ICD	87.87	87.67	88.24	87.83	81.82	80.67	85.98	85.39
	VCD	87.37	87.67	83.70	83.83	82.87	82.50	84.65	84.67
	ALD ² (Ours)	90.00	90.33	89.76	89.50	85.57	85.33	88.44	88.39
InternVL-3.5-8B	Greedy	91.61	91.67	86.67	87.22	82.67	84.00	86.98	87.63
	DoLa	87.83	87.52	81.83	82.45	79.50	80.63	83.05	83.53
	ICD	91.25	91.33	85.95	85.83	83.64	82.33	86.94	86.50
	VCD	90.42	90.05	88.09	87.83	82.13	81.00	86.88	86.29
	ALD ² (Ours)	91.83	91.85	88.16	88.75	84.33	85.63	88.11	88.74



(a) MME benchmark results across three LVLMs based on five methods.

Comment #2.4:

The related work on hallucination mitigation appears incomplete. I recommend expanding the discussion to include more recent studies on hallucinations in LVLMs, particularly works focusing on grounding, faithfulness, and decoding-time mitigation. For example, FaithScore: Fine-grained Evaluations of Hallucinations in Large Vision-Language Models, Aligning large multimodal models with factually augmented rlhf, A Unified Hallucination Mitigation Framework for Large Vision-Language Models. Mmbench: Is your multi-modal model an all-around player? Fine-grained and Explanable Factuality Evaluation for Multimodal Summarization. FIFA: Unified Faithfulness Evaluation Framework for Text-to-Video and Video-to-Text Generation, Decide: Alleviating Hallucination in Large Vision-Language Models via Multi-View Multi-Path Reasoning, FGAIF: Aligning Large Vision-Language Models with Fine-grained AI Feedback. From pixels to tokens: Revisiting object hallucinations in large vision-language models.

Thank you for your suggestion. We have updated the Related Work section to include a detailed discussion of these suggested studies on hallucination mitigation. The relevant revised contexts are as follows:

5. Related work

5.1 Hallucinations Mitigation in LVLMs

Hallucination mitigation in LVLMs has attracted increasing attention due to the inherent difficulty of aligning visual and textual modalities. Prior studies mainly explore three directions (Huo et al., 2024). The first direction strengthens model robustness through instruction tuning or alignment with curated, visually grounded datasets (Lee et al., 2022; Gunjal et al., 2024; Liu et al., 2023), including approaches based on factually augmented reinforcement learning and fine-grained feedback to improve visual faithfulness. The second direction applies post-hoc correction or verification with auxiliary analysis modules to detect and suppress hallucinated content after generation (Manakul et al., 2023; Zhou et al., 2024a), often relying on external reasoning paths, multi-view analysis, or additional verification signals.

The third line of work adjusts the decoding process to mitigate hallucinations through contrastive, constraint-based, or uncertainty-aware strategies (Li et al., 2023b; Chuang et al., 2023; Liu et al., 2024b). Representative methods include visual contrastive decoding and its variants (e.g., VCD and Med-VCD) (Mahdavi et al., 2026; Leng et al., 2024), instruction contrastive decoding (ICD) (Wang et al., 2024b), dynamic correction decoding (DeCo) (Wang et al., 2024a), multi-view multi-path reasoning (Qu et al., 2025), and self-introspective decoding (SID) (Huo et al., 2024), which dynamically evaluates token importance during generation. These decoding-time approaches are lightweight and model-agnostic and have shown promising effectiveness in reducing hallucinations without retraining large models.

Despite their success, existing methods largely operate at the data level or the output distribution level and rarely examine noise patterns within the model itself, particularly the instability introduced by redundant visual tokens in shallow layers. Such shallow-layer noise can propagate through deeper layers during decoding and ultimately trigger hallucinated predictions. To address this overlooked factor, we propose ALD2, a decoding-time method that explicitly targets shallow-layer noise through adaptive visual token pruning and confidence-aware layer-wise fusion, enabling more stable and visually grounded generation.

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To Reviewer #3:

Attention: We have rearranged the order of some of your review concerns to group similar points together for a more coherent response. However, the total number of your concerns remains unchanged.

Comment #3.1:

The authors treat hallucination as a shallow-layer noise propagation problem and attempt to denoise internal representations by pruning visual tokens at a selected shallow layer, followed by multiplying the resulting "denoised" distribution with the final-layer distribution. However, the manuscript does not formalize this design choice or explain why multiplicative fusion is preferable to alternative strategies (e.g., interpolation, logit addition with temperature scaling, KL-regularized fusion). The manuscript needs an ablation study comparing multiplication versus addition and interpolation.

Comment #3.2:

Most existing visual token pruning methods primarily aim to accelerate inference, whereas ALD uses pruning to modify the output distribution in order to reduce hallucinations. However, it is unclear whether the observed performance gains stem mainly from token pruning itself or from the proposed adaptive layer-wise selection and multiplicative fusion. Additional experiments are needed to disentangle the contribution of each component.

Comment #3.3:

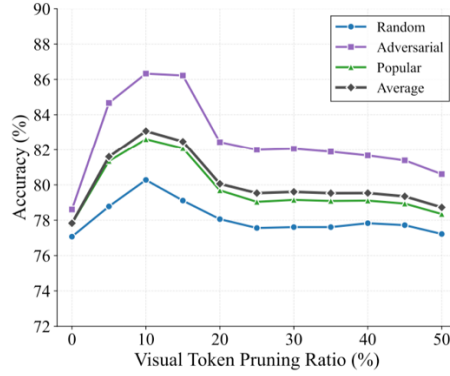
The ablation study should be strengthened by including the following analyses: Multiplicative fusion versus additive and interpolative fusion strategies; Prune-only decoding versus the full ALD² framework, to demonstrate that pruning alone does not account for the improvements; Sensitivity analysis of the layer search range (e.g., justification for restricting the search to the first three-quarters of the model depth), ideally supported by a quantitative plot.

Thank you for this insightful comment on identifying the source of our performance gains. Disentangling the contribution of token pruning from the adaptive selection and fusion mechanisms is essential for a clear understanding of ALD². In section 4.3. Ablation, we have demonstrated that multiplicative decoding only will get worse performance than greedy decoding. This could prove that visual token pruning is the key point to denoise and multiplicative decoding is used to amplify the effect of denoising and we also claim this point of view in our paper. The relevant contexts are as follows:

4.3. Ablation

...

Notably, when no visual tokens are pruned (i.e., pruning ratio set to 0), the resulting accuracy is lower than that obtained with greedy decoding, and that also verifies the importance of pruning redundant visual tokens. Furthermore, although we do not claim that a pruning ratio of 10% is universally optimal, our results indicate that pruning 10% of visual tokens is relatively effective.



For further addressing your concerns, we have added ablation experiments according to your suggestions. The revised contexts are as follows:

4.3. Ablation

...

Effect of Multiplicative Decoding and Visual Token Pruning. To further understand why ALD2 is effective, we compare the full ALD² framework with three variants: greedy decoding, pruning-only decoding, and a replacement of multiplicative fusion with additive fusion. As shown in Table 4, combining visual token pruning with multiplicative decoding consistently outperforms additive fusion, pruning-only decoding, and greedy decoding across random, popular, and adversarial settings. While pruning alone already improves performance over greedy decoding, indicating the effectiveness of suppressing redundant visual noise, its gains saturate without an explicit fusion mechanism. Replacing multiplication with addition leads to noticeable degradation, especially on popular and adversarial subsets, suggesting that additive fusion fails to sufficiently suppress residual noisy signals. In contrast, multiplicative decoding better emphasizes tokens jointly supported by both the denoised shallow layer distribution and the final layer distribution.

Table 4
Ablation study on fusion strategy and visual token pruning on POPE based on LLaVA-Next-8B.

Method	Random		Popular		Adversarial		Average	
	F1↑	Acc.↑	F1↑	Acc.↑	F1↑	Acc.↑	F1↑	Acc.↑
Greedy	86.87	87.50	84.07	84.33	82.26	82.17	84.40	84.67
Prune Only	90.00	90.33	85.00	86.24	81.62	78.83	85.54	85.13
Add + Prune	89.48	89.50	85.35	84.67	82.46	81.00	85.76	85.06
ALD²(Multiply + Prune)	90.00	90.33	89.76	89.50	85.57	85.33	88.44	88.39

Comment #3.4:

Although the authors report results on POPE and MME, these benchmarks are fundamentally yes/no evaluations. The manuscript does not clearly explain how model outputs are parsed into binary decisions. For example, how are responses with additional text, hedging, or ambiguous phrasing handled? Clarifying the answer extraction protocol is essential for ensuring fair and reproducible evaluation.

Thank you for your valuable feedback. We have pointed out in section 4.1 Experimental Setup that we apply Chain-of-Thought prompting when evaluating on POPE and MME benchmarks and that it can confirm LVLMS will generate the response corresponding to the shot. Consequently, the input will be like '<image> shot_question: Is there a/an <object> in this image? shot_answer: yes/no. <image> question: Is there a/an

<object> in this image? answer:’ and we guarantee that shot is not included in the test set. The relevant contexts from the manuscript are as follows:

4.1. Experimental Setup

...

For open-ended generation tasks, Chain-of-Thought prompting (Wei et al., 2022; Yin & Wang, 2025) is applied when evaluating POPE and MME, while CHAIR is evaluated using direct prompting.

...

Comment #3.5:

In Appendix A, the definitions of the CHAIR metrics appear inconsistent with the standard formulation. In particular, the denominators for instance-level and sentence-level metrics seem to be swapped or mislabeled. This should be carefully checked and corrected, or the authors should explicitly state that they follow the official CHAIR evaluation implementation.

We sincerely thank you for your careful reading and for spotting the inconsistency in our CHAIR metric definitions. We acknowledge that the expression of Chair metric in Appendix A was incorrect and has now been corrected. The revised contexts are as follows:

$$C_I = \frac{|\{all\ mentioned\ objects\}|}{|\{hallucinated\ objects\}|}$$
$$C_s = \frac{|\{captions\ with\ hallucinated\ objects\}|}{|\{all\ captions\}|}$$

Comment #3.6:

As stated in Section 3.2, the layer predictor is trained using prompt-layer pairs derived from ground-truth answers. This implies that ALD² is not a purely inference-only method, as it requires a supervised preparation stage. This distinction should be clearly acknowledged and discussed, especially when comparing ALD² to training-free decoding approaches.

We appreciate your highlighting the distinction regarding the training of the layer predictor. We have acknowledged this distinction in our new version. The revised contexts are as follows:

4. Experiments

4.1 Experiments Setup

Models and Baselines.... Notably, VCD, ICD, DoLa, and greedy decoding are all fully training-free methods, whereas ALD² introduces a lightweight classifier trained at the quantification stage to enable adaptive layer-wise denoising during decoding.

Comment #3.7:

Several recent hallucination mitigation methods are missing from the related work discussion, including Med-VCD, Dynamic Correction Decoding (DeCo), Self-Introspective Decoding (SID), and Language Contrastive Decoding (LCD). Incorporating these methods would provide a more comprehensive and up-to-date comparison with the current literature.

We thank you for bringing these relevant recent studies to our attention. We have updated the Related Work section to include a detailed discussion of these suggested studies . The relevant revised contexts are as follows:

5. Related work

5.1 Hallucinations Mitigation in LVLMs

Hallucination mitigation in LVLMs has attracted increasing attention due to the inherent difficulty of aligning visual and textual modalities. Prior studies mainly explore three directions (Huo et al., 2024). The first direction strengthens model robustness through instruction tuning or alignment with curated, visually grounded datasets (Lee et al., 2022; Gunjal et al., 2024; Liu et al., 2023), including approaches based on factually augmented reinforcement learning and fine-grained feedback to improve visual faithfulness. The second direction applies post-hoc correction or verification with auxiliary analysis modules to detect and suppress hallucinated content after generation (Manakul et al., 2023; Zhou et al., 2024a), often relying on external reasoning paths, multi-view analysis, or additional verification signals.

The third line of work adjusts the decoding process to mitigate hallucinations through contrastive, constraint-based, or uncertainty-aware strategies (Li et al., 2023b; Chuang et al., 2023; Liu et al., 2024b). Representative methods include visual contrastive decoding and its variants (e.g., VCD and Med-VCD) (Mahdavi et al., 2026; Leng et al., 2024), instruction contrastive decoding (ICD) (Wang et al., 2024b), dynamic correction decoding (DeCo) (Wang et al., 2024a), multi-view multi-path reasoning (Qu et al., 2025), and self-introspective decoding (SID) (Huo et al., 2024), which dynamically evaluates token importance during generation. These decoding-time approaches are lightweight and model-agnostic and have shown promising effectiveness in reducing hallucinations without retraining large models.

Despite their success, existing methods largely operate at the data level or the output distribution level and rarely examine noise patterns within the model itself, particularly the instability introduced by redundant visual tokens in shallow layers. Such shallow-layer noise can propagate through deeper layers during decoding and ultimately trigger hallucinated predictions. To address this overlooked factor, we propose ALD2, a decoding-time method that explicitly targets shallow-layer noise through adaptive visual token pruning and confidence-aware layer-wise fusion, enabling more stable and visually grounded generation.

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To Reviewer #4:

Research summary:

The research reported in this manuscript addresses the critical challenge of visual hallucinations in Large Vision-Language Models (LVLMs) by introducing a novel framework titled Adaptive Layer-wise Denoising Decoding (ALD2). The authors operate under the hypothesis that hallucinations are not solely semantic errors but are driven by noise propagation from shallow layers, where redundant visual tokens create unstable predictions. To counter this, the authors utilize a lightweight predictor to dynamically identify an optimal shallow layer, prune redundant visual tokens to suppress noise, and fuse the denoised distribution with the final layer's output via multiplicative decoding.

1. Mechanism of Action: The study establishes that while shallow layers often contain correct semantic signals, they are masked by high-entropy noise caused by visual redundancy. ALD2 mitigates this by pruning the bottom 10% of visual tokens (based on attention scores) in a targeted early layer and multiplying this "denoised" probability distribution with the final layer's distribution.
2. Performance Improvements: Extensive evaluations on benchmarks such as POPE, CHAIR, and MME across multiple models demonstrate that ALD2 consistently mitigates hallucinations.
3. Comparative Advantage: The method outperforms standard baselines, including Greedy Decoding, DoLa, Visual Contrastive Decoding, and Instruction Contrastive Decoding across tested metrics. For example, on CHAIR metrics, InstructBLIP-Vicuna-7B with ALD2 surpassed all baselines in reducing hallucinated objects.

Major Strengths: The major strengths of the research are:

1. The study shifts the focus from external input perturbations to internal model dynamics, effectively reframing hallucination as a layer-wise noise propagation problem where correct signals in shallow layers are masked by redundant visual tokens.
2. The proposed framework is rigorously tested across three representative LVLMs (LLaVA-1.5, InstructBLIP, and LLaVA-Next) and multiple diverse benchmarks (CHAIR, POPE, and MME), demonstrating consistent and significant accuracy improvements compared to strong baselines like VCD and ICD.
3. The ALD2 framework combines dynamic layer selection, visual token pruning, and multiplicative decoding to suppress noise while preserving deep-layer semantic reliability.
4. The manuscript has been improved by incorporating explicit research objectives and a dedicated section on theoretical and practical implications, directly addressing editor concerns regarding clarity and impact.

We thank the reviewer for the accurate summary and strong endorsement of our work. Your recognition of our 'noise propagation hypothesis' and the performance of the ALD² framework across various benchmarks is highly encouraging. Your comments not only validate the effectiveness of our method but also acknowledge our efforts in refining the manuscript's structure.

Major Weaknesses: The major weaknesses of the research are:

Comment #4.1:

While the paper acknowledges the increased latency, the discussion regarding the trade-off between this computational cost and the performance gains could be slightly expanded to better address practical deployment scenarios.

We appreciate your suggestion to expand the discussion on the trade-off between computational cost and performance gains. To address your concerns, we add the discussion about this trade-off in section 6 Theoretical and Practical Implications as follows:

6. Theoretical and Practical Implications

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From a practical perspective, the proposed ALD² framework offers a lightweight yet effective solution for reducing hallucinations in LVLMs, operating entirely at the decoding stage and easily integrating into existing pipelines. Its adaptive layer selection mechanism enables dynamic noise suppression tailored to different inputs, making it suitable for real-world deployment scenarios where reliability is critical. In practice, we believe the trade-off between increased decoding latency and improved trustworthiness is acceptable in settings where hallucination reduction and reasoning reliability are prioritized over strict efficiency constraints, such as offline evaluation, safety-critical decision support, medical or scientific analysis, and high-stakes human-in-the-loop applications. Conversely, for latency-sensitive scenarios like real-time interactive systems, greedy decoding or other lightweight strategies may be preferable. These considerations underscore that layer-wise denoising during decoding is both theoretically grounded and practically effective.

Comment #4.2:

The method relies on a fixed 10% ratio for visual token pruning, which is shown to be effective on average; however, a brief discussion on the potential benefits (or lack thereof) of a dynamic pruning threshold could further strengthen the "Limitations" section.

Thank you for this thoughtful suggestion. Discussing the potential of a dynamic pruning threshold versus our fixed ratio is an excellent addition that further clarifies the scope and future possibilities of our method.

Therefore, we revise the section Limitations as follows:

Limitations

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- ALD² denoises shallow layers by pruning a fixed proportion of visual tokens based on attention scores. This design choice favors simplicity and model-agnostic deployment, but it does not explicitly account for more complex vision-aware internal dynamics, such as adaptive token selection, vision-aware head behaviors, task-specific hallucination mitigation, or in-context visual reasoning (Zhou et al., 2025a; He et al., 2025; Zhou et al., 2024b). More precise token pruning strategies or dynamically adjusted pruning ratios may further improve denoising effectiveness. For example, Huo et al. (2024) proposed Self-Introspective Decoding (SID), which dynamically evaluates the importance of vision tokens during decoding to adaptively reduce hallucinations in large vision-language models.

Comment #4.3:

The explanation for the counter-intuitive result where the LLaVA-Next-8B model achieves high performance gains despite low predictor accuracy (around 30%) could be deepened to better illustrate the

robustness of the layer selection mechanism.

We are grateful for your keen observation regarding the counter-intuitive results on LLaVA-Next-8B. We add the detailed explanation about this divergence. The relevant revised contexts are as follows:

4.2.3 Accuracy of predictor

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Furthermore, it can be observed that LLaVA-Next-8B achieves high performance gains despite low predictor accuracy. This discrepancy can be attributed to the objective used during the quantification stage. The layer predictor is trained to select the layer that maximizes the probability of the ground-truth token after layer-wise multiplicative fusion, rather than to directly predict token correctness. As a result, probability maximization does not necessarily guarantee correct token generation.

Additionally, according to Figure 9, a larger proportion of its layers exhibit high visual noise, making the separation between informative and noisy layers less distinct. Although the predictor achieves lower accuracy, correctly predicted cases almost always yield the desired token after denoising, and some cases predicted incorrectly even produce correct outputs due to effective suppression of shallow-layer noise.

In contrast, LLaVA-1.5-7B and InstructBLIP-Vicuna-7B show substantially reduced noise in deeper layers. Higher layer-selection accuracy does not always translate into correct token generation, as remaining errors are often dominated by semantic ambiguity rather than visual noise. These observations indicate that layer prediction accuracy and generation improvement are not strictly correlated.

Grammar and Readability: The manuscript is well-written, and the revisions have significantly improved its overall clarity and flow. The English is coherent and suitable for an academic audience. The figures and tables are clear and informative. A final round of minor proofreading is recommended to catch any lingering typographical errors.

We sincerely appreciate your positive feedback on the improved clarity and readability of our manuscript. Following your recommendation, we have conducted a final, thorough proofreading to ensure the paper is free of typographical errors.

Specific Comments: My specific comments concerning this manuscript are:

Comment #4.4:

The manuscript notes a 94% increase in latency compared to greedy decoding. While this is acknowledged, it would be beneficial to add a brief discussion in the "limitations" or "conclusion" section regarding potential strategies to optimize this in future work, or to more explicitly frame the trade-off for applications where accuracy is paramount over speed.

We appreciate you highlighting the need to discuss potential optimization strategies. We discuss the potential strategy to optimize in our revised version. The relevant revised contexts are as follows:

4.3. Ablation

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Regarding scalability, while our evaluation focuses on standard-length text responses and moderate-resolution visual inputs, the relative latency overhead of ALD² may grow for longer responses or higher-resolution images. Since both the additional layer-wise inference and token processing scale with the number of tokens and visual features, scenarios with extended dialogues or dense visual inputs could experience further latency increases. Notably, the latency can be reduced by leveraging our visual token pruning strategy, where less informative tokens are directly removed from the computation graph rather than merely

zeroing their hidden states. This reduces memory usage and computation in subsequent layers, offering a practical avenue to mitigate latency while retaining most of the accuracy improvements.

Comment #4.5:

The divergence between the low predictor accuracy (30%) and the high performance gains on the LLaVA-Next-8B model is a compelling finding. I suggest expanding the explanation to hypothesize why this specific model benefits so significantly even when the "optimal" layer is not perfectly predicted. This would add depth to the analysis of the method's robustness.

Thank you for finding the divergence between predictor accuracy and performance compelling. We have provided a detailed response under Comment 4.3.

Comment #4.6:

The choice of a fixed 10% pruning ratio is supported by the ablation study. A short sentence could be added to discuss whether the sensitivity to noise varies significantly across different layers, which could inform future adaptive pruning strategies.

We appreciate this insightful suggestion. We discuss this in section 2.1 Noise across Different Layers of LVLMs, and we use entropy to represent noise of each layer. The relevant contexts are as follows:

2.1 Noise across Different Layers of LVLMs

Inspired by noise suppression across decoding layers in LLMs (Zhou et al., 2025b), we investigate vertical noise propagation in LVLMs. Noise refers to task-irrelevant interference introduced by redundant visual tokens, which disrupt attention allocation and cause the model to fluctuate among unrelated candidates, producing high-entropy probability distributions. Lower entropy indicates higher confidence and reduced noise. To analyze this phenomenon, we prune redundant visual tokens at shallow layers and propagate the pruned representations to obtain a denoised distribution. Figure 3 shows a representative example (additional cases appear in Appendix B). Shallow layers exhibit high entropy (around 10), reflecting substantial noise and uncertainty, while deeper layers display much lower entropy (around 1), indicating more stable representations. After pruning, the resulting shallow-layer denoised distribution ("Denoised" row in Figure 3) maintains consistently lower entropy than the final layer. Pruning removes redundant visual information, suppresses shallow-layer noise, and yields a probability distribution more concentrated on the target token.

This aligns with the observation in Figure 2, where we inspect the distribution immediately before decoding the final answer. In the final layer, the model hesitates between "yes" and "no" because their probabilities are nearly identical. This ambiguity originates from shallow-layer noise, where the probability of "yes" is much higher than that of "no", injecting uncertainty into the final-layer prediction. After denoising, the model can confidently select the correct token.

Input image :



Question:

Please help me describe the image in detail.

LVLm response:

The scene features a baseball game taking place on a field, with several players involved in various positions.

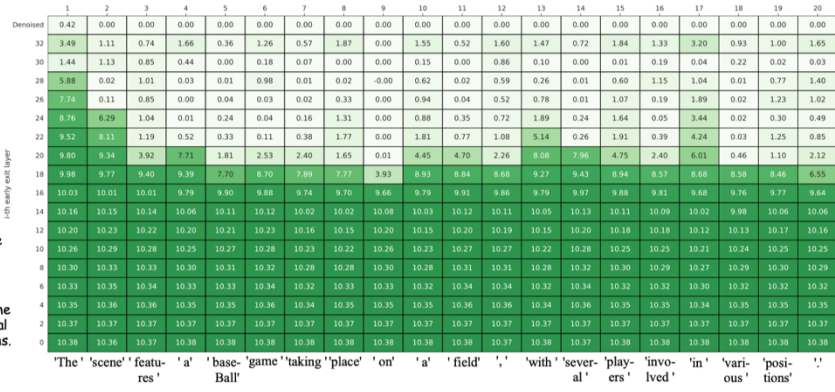


Figure 3: The entropy of the probability distribution for the final layer and several early layers in InstructBLIP-Vicuna-7B under teacher forcing at each step. The horizontal axis represents the decoding step, while the vertical axis (ranging from 0 to 32 in intervals of 2) denotes the considered early layers and the final layer. For clearer illustration, we present the first 20 tokens produced by the model.

Concluding Remarks: The authors have done a commendable job revising the manuscript, particularly by adding clear research objectives and a dedicated discussion on theoretical and practical implications. The proposed ALD2 framework presents a novel and effective solution for hallucination mitigation in LVLms, supported by experimental results. I recommend that the authors address the minor comments regarding the computational latency trade-offs and the predictor's behavior on specific models to further polish the paper. Thank you for your high praise of the ALD² framework and your recommendation. We value your final suggestions and have addressed the points regarding latency trade-offs and predictor behavior to further polish the paper.